

Ayush Kumar

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EDUCATION

Tomsk State University, Tomsk, Russia

Specialization (5 years) – **Computational Mathematics and Computer Modeling**

Faculty of Mathematics and Computer Science

EXPERIENCE

T-Bank Education

April 2025

AI Apprenticeship

- Developed and fine-tuned an **LLM-based question-answering system** that **improved automated support response accuracy by 25%** and **reduced manual ticket handling by 15%**.
- Collaborated with senior engineers to implement a new evaluation framework, **increasing model F1-score by 10%** and improving response relevance through rigorous **data preprocessing**.

Avito

Machine Learning Intern

May 2025

- Collaborated on a **model performance optimization** project, focusing on improving **prediction accuracy** through data preprocessing and validation.
- Supported **deployment of ML pipelines** using **PyTorch** and **Scikit-learn** for production-grade tasks.

PROJECTS

AuthCore – Custom Authentication & RBAC System

- AuthCore - Custom Authentication and Role-Based Access Control (RBAC) System
- Developed a secure and stateless authentication system in **Python**, leveraging **JSON Web Tokens (JWT)** for efficient session management.
- Implemented robust password security using **bcrypt** for hashing.
- Designed and built a fine-grained **Role-Based Access Control (RBAC)** model to safeguard **RESTful API** endpoints.
- Created custom middleware to validate user permissions against a **PostgreSQL** database.
- Automated deployment and maintenance workflows through Shell scripting.
- Delivered a **scalable, efficient, and secure backend solution** tailored to meet modern application security needs.

Code-Gym-2025-Q-A

Award-Winning Solutions for the T-Bank Code Gym Programming Competition

- Secured 3rd place** in a high-pressure programming competition by developing robust solutions for 10 diverse challenges involving logic, algorithms, and mathematical modeling.
- Implemented advanced algorithms, including **dynamic programming** for combinatorial counting, **graph traversal (BFS)** for pathfinding, and **binary search** for optimization tasks, all under strict performance constraints.
- Engineered highly-optimized solutions in C++ and Python by applying concepts from **number theory** (modular arithmetic, bitwise operations) and **probability** (expected value), and by modeling complex scenarios with **systems of linear equations**.

CERTIFICATIONS

- [Deep Learning AI Specialization](#) – DeepLearning.AI (January 2024)
- [Neural Networks and Deep Learning](#) – DeepLearning.AI (December 2023)
- [IBM Machine Learning](#) – Coursera (July 2023)
- [Python Pro Bootcamp \(100 Days of Code\)](#) – Udemy (December 2022)
- [Full-Stack Web Development Bootcamp](#) – Udemy (May 2022)

SKILLS

- Machine Learning:** Deep Learning, LLMs, CNNs, NLP, Data Preprocessing
- Frameworks & Tools:** PyTorch, TensorFlow, LangChain, Scikit-learn, Pandas
- Programming:** Python, SQL, Java, C++
- Concepts:** Model Evaluation & Validation, Feature Engineering, Hyperparameter Tuning, Prompt Engineering, Transfer Learning & Fine-Tuning, Algorithm Design & Analysis, MLOps Principles, System Design

ADDITIONAL

- Languages:** English (Fluent), Russian (Intermediate)
- Awards:** **3rd Place Winner**(Siberian Region Russia), Code Gym by Т-Банк Образование (April 2025); AvitoTech ML CUP 2025 by Avito hosted by Open Data Science(ods.ai) **Awarded Diploma for Successful Challenge Completion**.